Final Report

Team 82

1 The Solution

PaisaPath is an offline-first progressive web app that teaches financial literacy to Grade 11–12 students by having them *live* a fictional character's money year rather than read rules about it. The problem statement, *Empowering the Next Generation Through Financial Literacy*, asks for a scalable way to cover five areas — banking and digital payments, budgeting, saving and borrowing, investing, and fraud safety — for a pilot of about 1,500 students in Hyderabad, delivered through NGO partners.

Our reading of the problem has not changed since M1: money lessons fail because consequences arrive weeks later and cost too much to learn from. So the app makes consequences immediate, cheap and reversible. The main mode is **Paisapur**, a top-down tile town in the style of a classic handheld RPG. The student walks the town, enters buildings, and plays a prologue plus six chapters — bank, budget, save, borrow, invest, protect — each closing with a "master's challenge" gate. The character can overspend, fall for a simulated scam or miss a goal; the student can then rewind to an earlier decision through the **What-If Machine**, choose differently and compare the two lives. Both timelines replay from the same seed, so any difference comes from the decision and not from luck. A rival character, re-simulated from that same seed, gives a constant side-by-side contrast.

The setting is deliberately hard: low-end and often shared phones, unreliable classroom internet, four languages, and facilitators who need to see where a class is stuck. Because the users are minors, the app never asks for or stores real money, family or message data — a class code and a picture PIN are the only identity — and it never names or recommends a real financial product.

2 Approach and Architecture

The app is a React 19 / TypeScript / Vite PWA. The architecture is organised around one rule: **the simulation is pure and the content is data.**

- `core/` — a pure, deterministic TypeScript simulation: the money engine, a seeded fictional market, world/collision maps, a story-script interpreter, rival simulation, and the timeline (checkpoint, rewind, compare). It imports no React, no database and no network. Clocks and randomness are injected, so the same seed and the same choices always produce the same state. This is what makes rewind and replay trustworthy rather than approximate.
- `platform/` — the only code allowed to touch IndexedDB, the network or the backend: saves, snapshots, profiles, the telemetry outbox, sync status and translation. Everything else imports from one barrel, so storage can change without touching gameplay.
- `content/` — 395 JSON files: story scripts, world maps, lessons, glossaries, quizzes, scenarios, the concept registry and the four language bundles. New scenarios need no code.
- `ui/` — a presentational pixel/tile design kit shared by every screen.

These boundaries are not conventions but **lint-enforced** import rules, so a pull request that reaches into the database from game logic fails CI. A standalone content validator (no dependencies, runs from the repository root) independently checks IDs, effect paths, story steps, warps, NPC scripts and translation key coverage.

Story scripts are JSON step lists (`say`, `choice`, `if`, `do`, `quiz`, `concept`, `minigame`, ...) with conditions written as strings such as `chapter>=2 && !flag:x`. A `minigame` step can stage a scene, convert a game's result back into story money, and run follow-up dialogue for a choice made inside the game — which is how the mini-games stay inside one economy instead of being a separate arcade bolted on.

Since M2 the three parallel branches have been **merged into a single app**. The investing rooms and the Stay Safe simulators are now reachable from inside the town and cross-linked by concept, so a student who stumbles on `loan_app_trap` in the story is pointed at the simulator that drills it.

3 Features

- **Paisapur (main mode).** Prologue and six chapters, 53 story scripts across 10 files, a walkable town with shops, a bank, a lender and NPCs, a seeded market, needs/wants/goal economy, badges, and the What-If

Machine with a rival timeline.

- **Mini-games.** 15 Everyday Money games (document hunt, bank counter, jar split, goal builder, spending detective, surprise month, debt ladder, ...), 9 investing modules and 11 Stay Safe simulators — 52 registered IDs in total. Games read money only through shared context helpers, and a coherence test fails the build if a game and the story disagree about income, bills or prices.
- **Investing.** Task-based rooms with typed answers, hints and points; interactive labs; a Time Machine that runs 500 simulated futures and reports a *range* with its assumptions, never a single number; a 12-week stock-market simulator with six fictional companies and a coach that never says what to buy; a mutual-fund challenge; and an investing/gambling/scam sorter.
- **Stay Safe.** A simulated phone with SMS, chat, email, calls and search; a Scam Lens; a UPI simulator (pay by QR, then a fake collect request); Scam Duels; XP, stars, ranks and badges.
- **Facilitator dashboard.** Classes, unlocks, and an Insights tab that flags concepts a class finds hard (below 50% with at least 6 answers), fed by pseudonymous events. Row-level security means a facilitator sees only their own classes. Students never log in to the backend.
- **Offline and sync.** Installable PWA; saves and snapshots in IndexedDB; a telemetry outbox that uploads in batches of at most 200 with back-off and never sends the same event twice; class unlocks flow back down.
- **Languages and accessibility.** Four languages (English, Telugu, Hindi, Urdu) with per-key fallback to English, plurals via `Intl.PluralRules`, rupee formatting with Indian digit grouping, and right-to-left layout for Urdu. Language packs load on demand rather than all at boot. A `timersOff` setting makes every timer visible but non-expiring.

4 Results

All figures below come from the merged `main` branch, not from the separate branches reported in M2.

Check	Result	What it covers
Test suite (Vitest)	**1703 / 1703** pass, 85 files	money engine, market, story content, rewind, duels, all mini-games, sync outbox, facilitator insights, translation
Types (`tsc -b`)	clean	strict mode across 452 source files (≈37.9k lines)
Lint (oxlint)	0 errors	includes the folder-boundary import rules
Content validator	**PASS, 0 errors**	137 concepts, 81 misconceptions, 93 scenario files, 53 story scripts, 52 game IDs, 6799 English keys, 3812 referenced keys
Brand-name guardrail	502 checks pass	no real bank, fund, app or company name appears in any lesson, game or scenario text
Investing disclaimers	12 checks pass	every investing route shows the "simulation, not advice" banner; no screen shows a lone projected number
Colour contrast	23 checks pass	every text/background token pair $\geq$ 4.5:1 (WCAG AA)

Determinism. The same seed produces the same market and the same story outcome. A bot plays the entire year through all six chapters as part of the suite, and rewinding leaves the original timeline unchanged. Every number a lesson room quotes is tested against the shared assumptions file, so the prose and the maths cannot drift apart.

Fairness of the fund model. The mutual-fund engine is tested so that active and sector funds grow about the same as the index fund after cost, and no text calls any fund best — the lesson is about cost, not about picking a winner.

Translation. This was M2's largest outstanding gap: the Telugu, Hindi and Urdu folders were empty. They are now populated and verified against English key-by-key.

Language	Keys translated	Coverage	Complete namespaces
Hindi	5716 / 6799	84.1%	all of Paisapur, Stay Safe, Scam Duels, arcade, setup
Urdu	5236 / 6799	77.0%	as above, right-to-left verified
Telugu	4770 / 6799	70.2%	as above

57 of the 82 string namespaces are complete in all four languages, and **Paisapur itself — the main mode — is 100% complete in all four** (1456 keys each, 4368 translations). In total 15,722 of 20,397 non-English

strings are translated. What remains is standalone lesson bundles (bank, borrow, save, two planning lessons, the Time Machine, and some Telugu and Urdu investing rooms); none of these sit on the Paisapur story path, and a missing key falls back to English rather than breaking the screen.

5 Limitations

We would rather state these than let a demo imply otherwise.

- **Lesson bundles are not fully translated** (figures above). The main mode is; the standalone lessons partly are.
- **Language is chosen only at first-launch setup.** There is no in-app switcher yet, so changing language means re-running setup.
- **No real-device or learning-gain data.** We have not yet done the 360×640 pass on a low-end phone, nor the pre-check/post-check measurement that would show whether any of this actually teaches. Timing tests are load-sensitive, so we do not claim a hard performance bound on low-end hardware.
- **Offline is unit-tested, not field-tested.** The outbox and PWA state are covered by tests, but a full airplane-mode-then-reconnect run on a phone is still pending.
- **"Listen" uses the browser voice**, not recorded audio, so pronunciation quality varies by device.